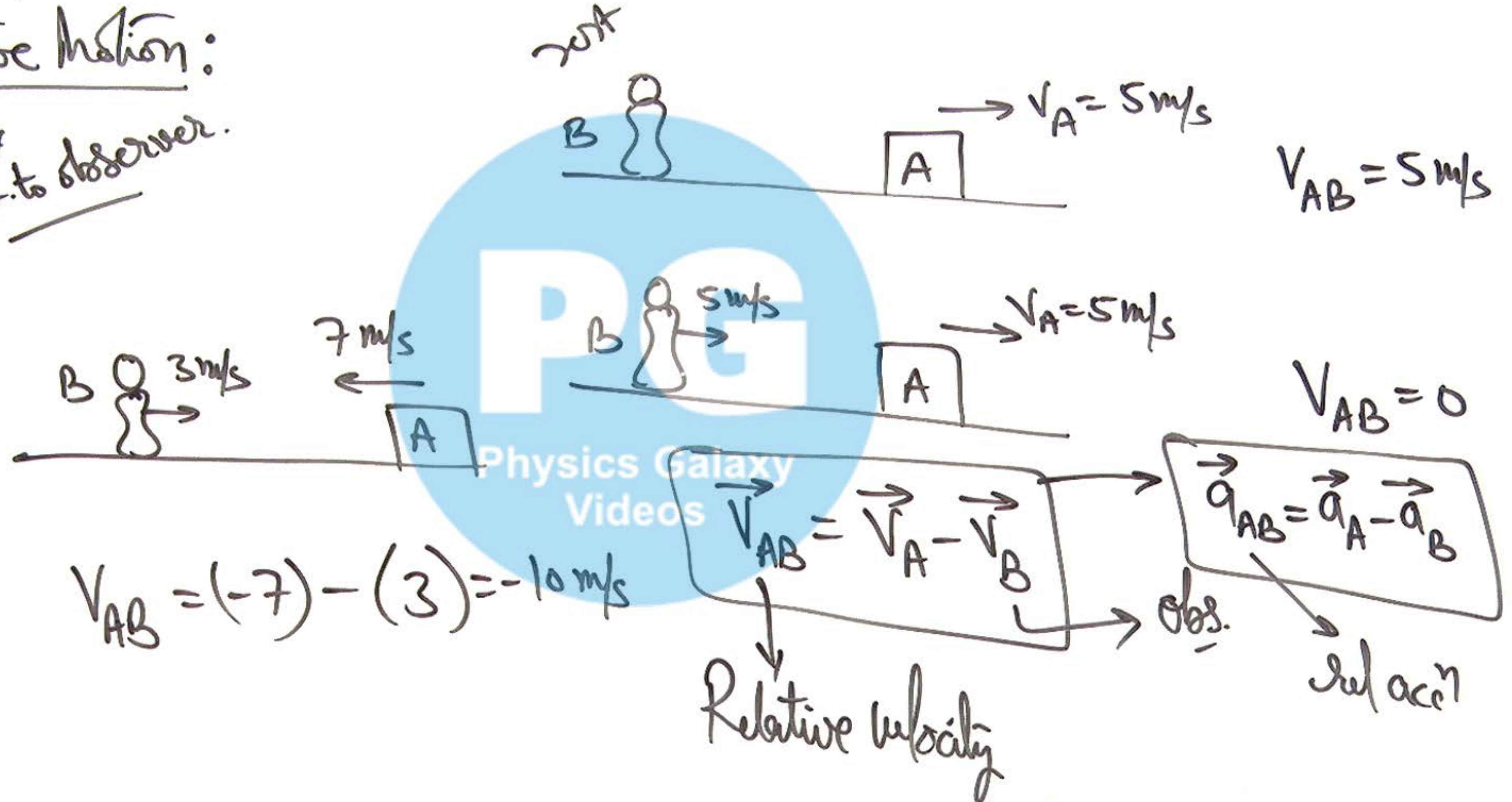


① Reference frame

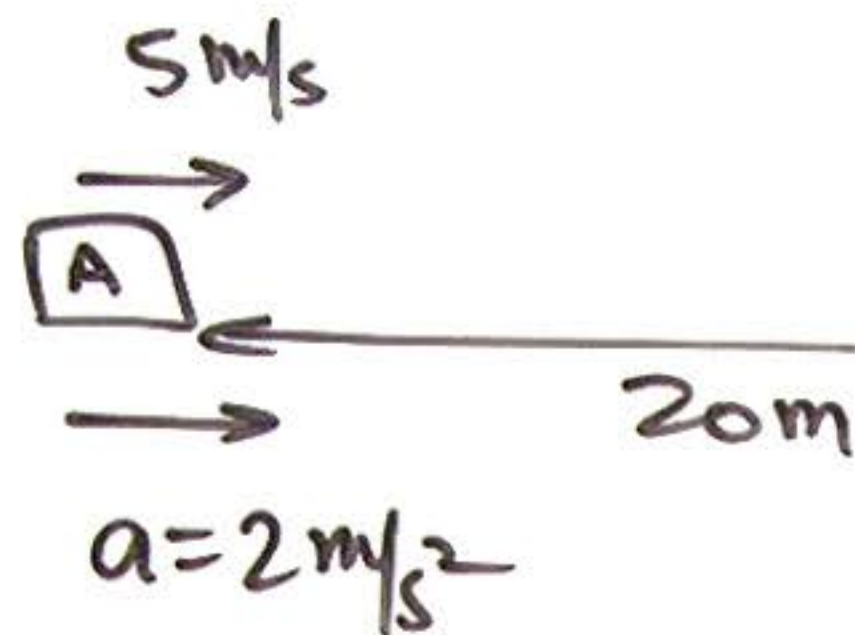
& Relative Motion:

↓
w.r.t. to observer.



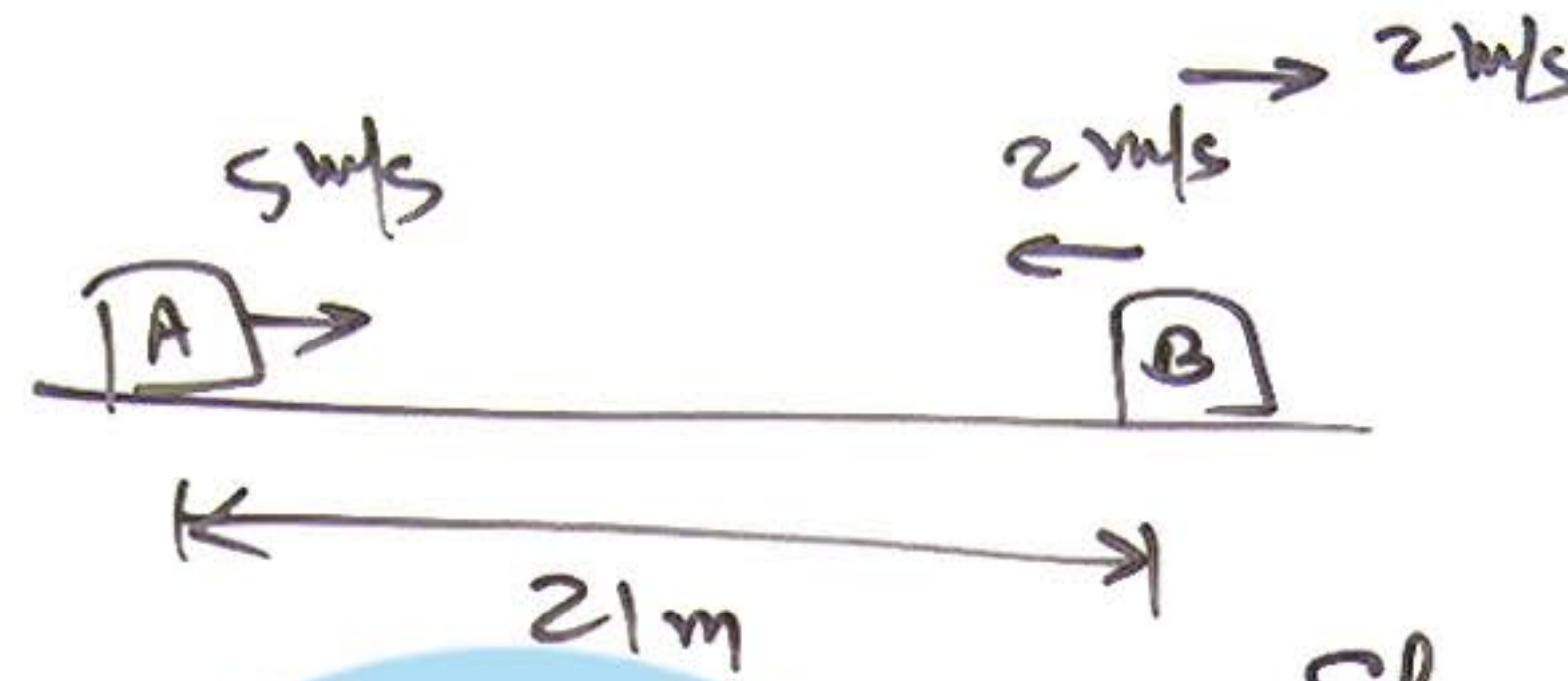
① Reference frame
& Relative Motion:

find time
when A
overtakes B.



$$V_{AB} = 5 - 10 = -5 \text{ m/s}$$

$$a_{AB} = 2 \text{ m/s}^2 - 0 = 2 \text{ m/s}^2$$



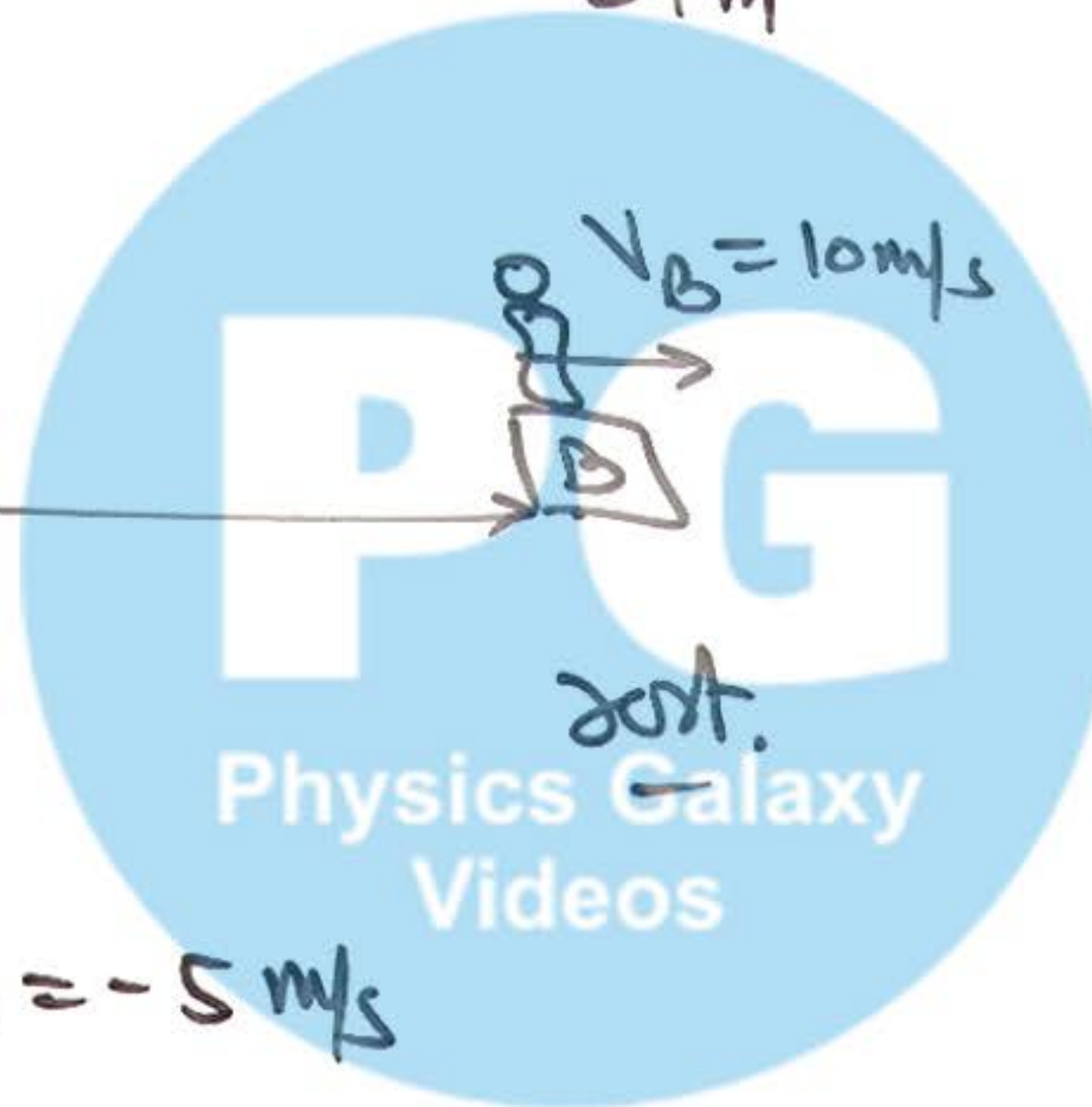
$$t_1 = \frac{s}{v} = \frac{21}{7} = 3 \text{ s}$$

Speed eqⁿ $t_2 = \frac{21}{3} = 7 \text{ s}$

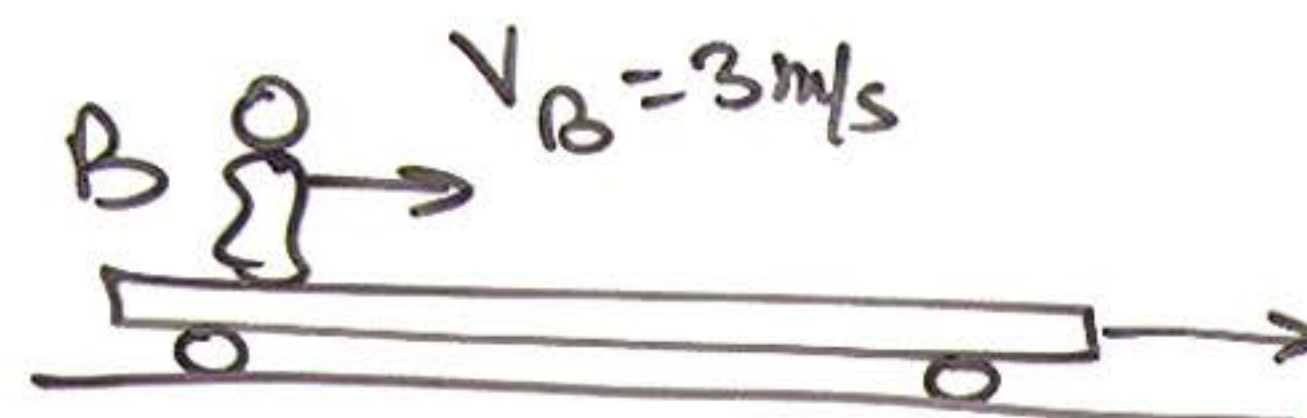
$$s = ut + \frac{1}{2}at^2$$

$$+20 = -5t + \frac{1}{2}(2)t^2$$

$$t = \text{---} \text{ Ans}$$



② Motion in a moving frame:

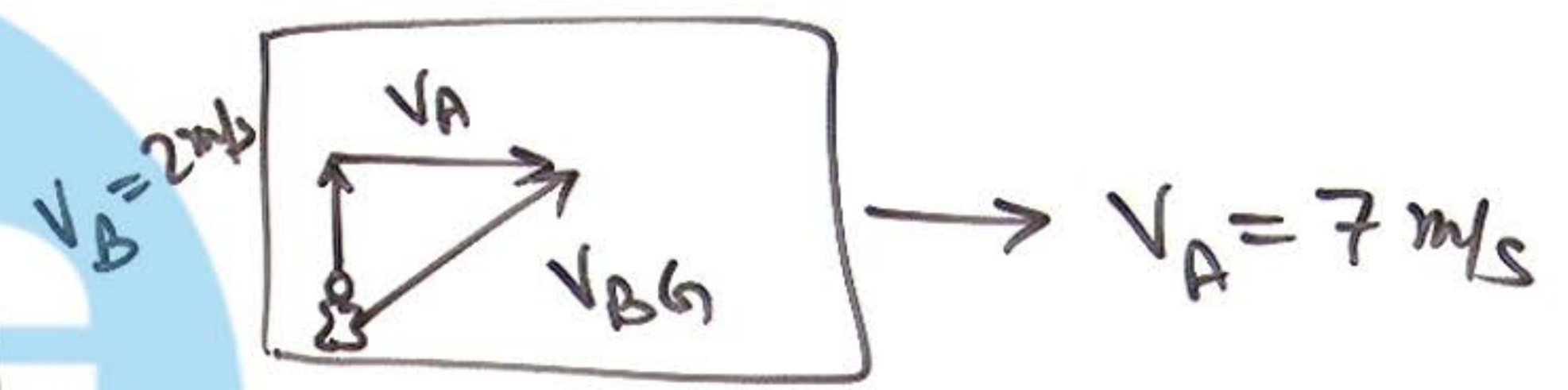


$$V_{BG} = 3 + 5 = 8 \text{ m/s}$$

✓ $\vec{V}_{BG} = \vec{V}_B + \vec{V}_A$

body vel

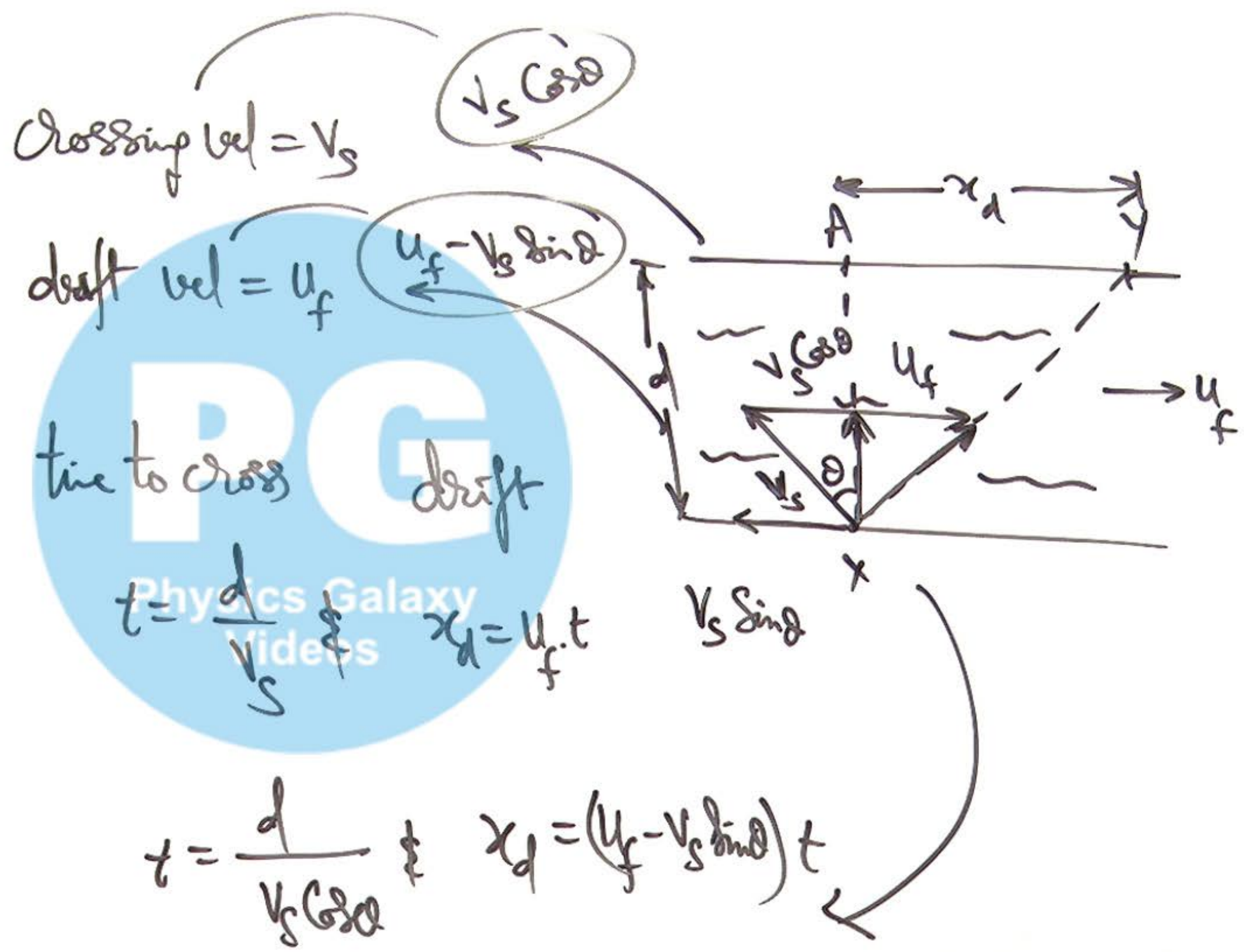
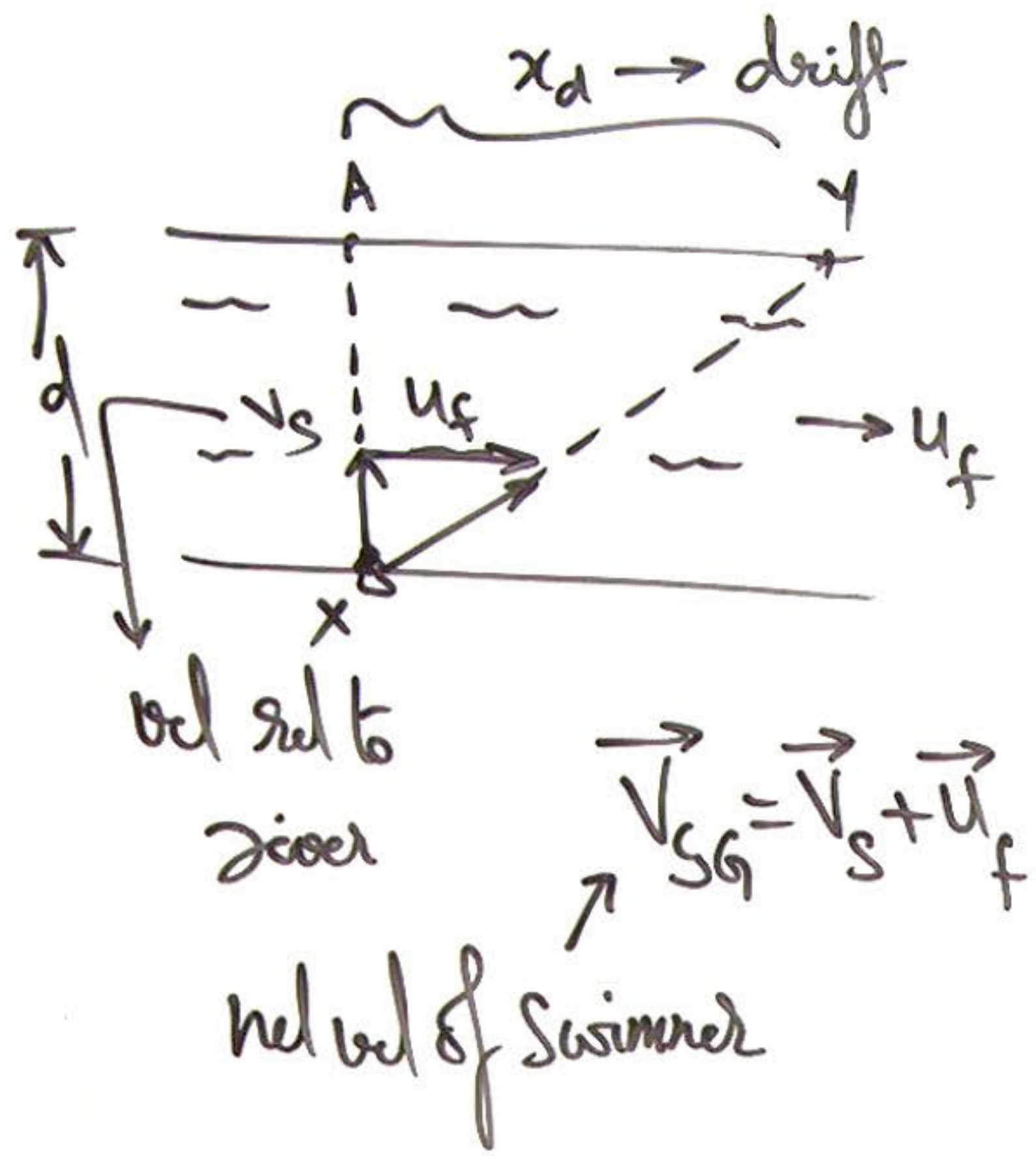
frame velocity
in which V_B is given



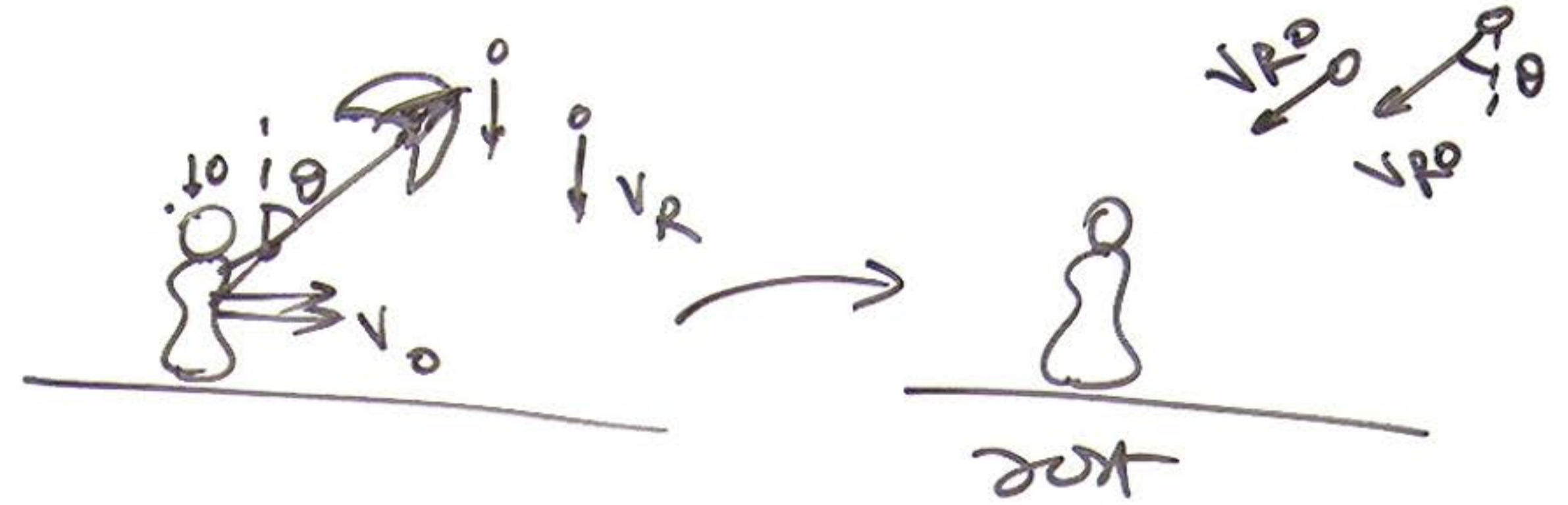
$$\vec{V}_{BG} = \vec{V}_B + \vec{V}_A$$



③ River flow cases:



④ Rain Fall Cases:



Vel of rain drops

w.r. to a moving observer

$$\vec{V}_{RO} = \vec{V}_R - \vec{V}_O$$

$$\vec{V}_{RO} = \vec{V}_R - \vec{V}_O$$



$$\theta = \tan^{-1}\left(\frac{V_O}{V_R}\right)$$

Physics Galaxy
Videos